

DATA WAREHOUSE - DATA mining

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Assessment Tool - 01

Q.

City Hospital

Data warehouse

→ snowflake schema

Patient - Dim

Doctor - Specialization - Doctor - Dim - Department - Dim - Hospital

Fact - Visit

Patient - ID

Doctor - ID

Department - ID

Time - ID

Visit - ID

Avg - ID

Time - Dim

Day → month → Quarter → year.

⑥ OLAP Operations:-

Basic cuboid: [Patient, Doctor, Department, Time]

Operations:-

* Roll up Time → year

* Roll up Patient → All

* Roll up Doctor → All

* keep Department dimension unchanged

* Aggregate Avg. charge

Output:-

[Department | year | Average charge]

① Number of cuboids

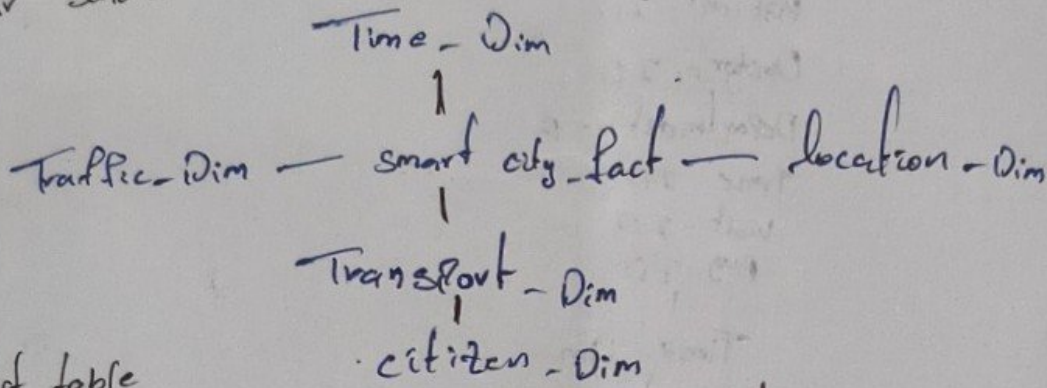
Formula Total cuboids = Product of hierarchy levels

Each dimensions = 4

$$4^4 = 256$$

2 Smart City Star Schema

Star Schema



Fact table

measures

vehicle count

congestion level

Energy consumption

Air Quality index

Noise level

Ridership

Productivity

Time

year

month

Day

Hour

location

City

Zone

area

traffic

Sensor ID

Road type

Transport

Bus

metro

Route

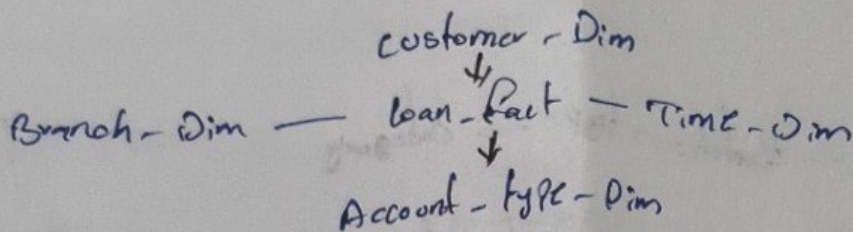
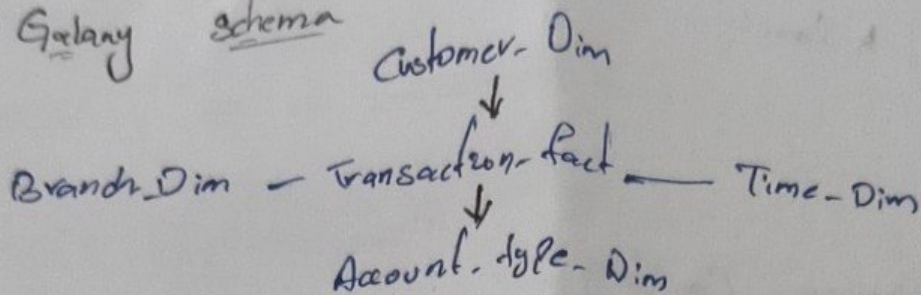
Justification

- measures store numerical values
- Dimensions enable multidimensional analysis by location, time, transport, and traffic.

Q₃

Banking Data Warehouse

② Galaxy schema



Transaction fact

Transaction count

Amount

Loan fact

Loan Amount

Interest

③ OLAP operation

Starting cuboid customer, Branch, Time, Account-type.

Operations

- Roll up Time → year
- Roll up customer → All
- Roll up Account type → All

• keep ~~Branch~~

• Aggregate Average Amount and Loan Amount.

Output:

| Branch | Year | Avg transaction | Avg loan |

(C) Cuboids
 4 dimensions
 each has 4 levels
 $4^4 = 256$

24 Manufacturing Company

Roll-up

Roll-up product: Batch $\rightarrow$ type $\rightarrow$ category

Roll-up factory: Line $\rightarrow$ Department $\rightarrow$ factory

Keep current year

Output: Average defect rate by product category.

Drill-down
factory

↓
Department

Product
category

↓
type

Analysis defect rate

slice

condition: Supplier Rating ≥ 4

Display: production unit

Dice:

select

- product categories
- factory location

measures

- Defect Rate
- Supplier Reliability %

Q5 Retail chain

Roll-up

Roll-up : sku $\rightarrow$ subcategory $\rightarrow$ category

Time : Day $\rightarrow$ month $\rightarrow$ Quarter

measure : Inventory Turnover

Drill-down

Region

↓

city

↓

store

Electronics

↓

subcategory

↓

sku

measure

Sales Revenue

slice:

condition:

Gender = male

membership = Bronze

measure:

Sales Revenue

Dice:

Selected

- cities
- Product categories
- Time Period

measure

- Sales Revenue
- units sold.